

REQUIREMENT ANALYSIS Technology stack (Architecture & Stack)

Date	FEBRUARY 2026
Team ID	LTVIP2026TMIDS58722
Project Name	ToyCraft tales: tableau's vision into toy manufacturer data
Maximum Marks	4 Marks

Technical Architecture

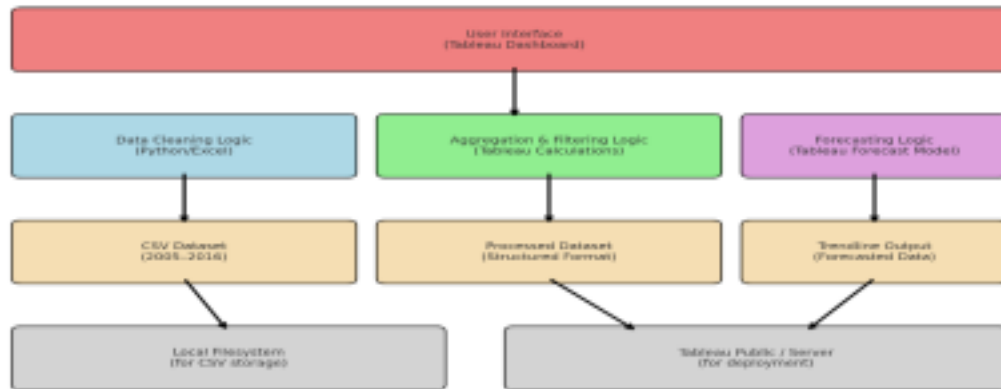


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Dashboard interface for users to interact with data	Tableau Public
2.	Application Logic-1	Data cleaning and transformation	Excel(pre-Tableau)
3.	Application Logic-2	Data aggregation by year, category, region	Tableau calculated fields
4.	Application Logic-3	Forecasting based on historical trends	Tableau Forecasting
5.	Database	CSV dataset with shipment and category info	Excel sheet

6.	Cloud Database	Not applicable	Tableau cloud
7.	File Storage	Upload and store toy dataset	Local drives or google drive
8.	External API-1	Weather data to correlate seasonality	Open WeatherAPI
9.	External API-2	Social media trend integration	Google Trends
10.	Machine Learning Model	Predictive modeling	Tableau's built-in forecast model
11.	Infrastructure (Server / Cloud)	Cloud-hosted dashboard viewable by users	Tableau Server/Tableau public

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Python (data cleaning)	Python
2.	Security Implementations	Restricted access via Tableau login	IAM (Tableau server)
3.	Scalable Architecture	Tableau scales to multiple dashboards/users without code changes	Tableau cloud Architecture

4.	Availability	Dashboard hosted on Tableau Public with 24/7 access	Tableau server/Tableau public
5.	Performance	Optimized visual queries, aggregated filters, and trendline calculations	Tableau filtering